

Ian Pamerleau | Postdoctoral Researcher

Contact Information

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Earth and Atmospheric Sciences
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Research Interests

Geophysics, Numerical Modeling, Volcanos, Topographic Analysis, Icy Worlds, Hazards

Education

2021–2026	PhD	Purdue University, Planetary Science Advisor: Michael Sori
2021	BS	University of Pittsburgh, Mathematics
2021	BS	University of Pittsburgh, Geology

Professional Experience

2026–Present	Postdoctoral Researcher, Cornell University (Advisor: Matthew Pritchard)
2025	NASA JPL Planetary Science Summer School: TIDES Team
2021–2026	Graduate Research Assistant, Purdue University
2019–2021	Research Assistant, University of Pittsburgh (Advisor: Eitan Shelef)
2019	Research Assistant, University of Pittsburgh (Advisor: Ming Chen)

Awards & Honors

2026	Purdue EAPS Department Outstanding Graduate Student Award
2025	Pellas-Ryder Award
2025	Darrell Leap Hyrdogeology Purdue Graduate Research Award
2024	Purdue Graduate Student EXPO Award: Outstanding Poster
2022	NSF Graduate Research Fellowship Program Honorable Mention
2022	Purdue Donald W. Levandowski Memorial Scholarship in Geology
2021–2025	Purdue Ross Fellowship
2021	University of Pittsburgh Flint Memorial Field Geology Award
2020	University of Pittsburgh Brackenridge Research Fellowship
2019	NASA Pennsylvania Space Grant Consortium Scholarship

Teaching Experience

2025	Teaching Assistant, Purdue University EAPS 309, Computer-Aided Analysis for Geosciences
2025	Teaching Assistant, Purdue University EAPS 111, Physical Geology Lab
2024	Teaching Assistant & Guest Lecturer, Purdue University, EAPS 556, Planetary Surface Processes
2024	Teaching Assistant, Purdue University, EAPS 507, Introduction to Analysis and Computing with Geoscience Data
2023	Guest Lecturer, Purdue University, EAPS 354, Earth and Planetary Geophysics
2020	Math Assistance Center Tutor, University of Pittsburgh, General University Level Math Including Calculus and Analysis

Field Work Experience

2024	Mars Desert Research Station, Crew 305, Executive Officer, Crew Geologist
2022	Field Training and Research Program at Meteor Crater
2021	Southern Utah University, GEO4960-30Y, Geology Field Camp (Online)

Leadership Experience

2024–2026	Purdue EAPS Graduate Student Association, President
2023–2024	Purdue EAPS Graduate Student Association, Treasurer
2021–2022	Purdue EAPS Graduate Student Association, Secretary
2020–2021	University of Pittsburgh Society of Physics Students, President
2018–2020	University of Pittsburgh Society of Physics Students, Treasurer
2018–2021	University of Pittsburgh Jazz Ensemble, President
2018–2019	University of Pittsburgh Fencing Team, President

Service

2026	Lead on AGU Workshop: Finite Element Method Modeling of Geophysical Processes for Earth and Planetary Science
2025–2026	Planetary Science Journal Reviewer
2023	Icarus Reviewer
2022–2023	Purdue EAPS Planetary Colloquium Coordinator

Publications

- [6] Miller, S. G., Alian, O., Borrelli, M., Christian, J., Detelich, C., Ferreira, J. P., Hon, O., Jhoti, E., Kubas, A., Lutz, K., **Pamerleau, I. F.**, Pickett, N. B., Rountree, R., Sacks, L. K., Thakar, B., Tuggle, J., Valantinas, A., Zomerkijk-Russell, S., Nash, A. E., Scully, J. E. C. (2026), A Triton Multiple-Flyby Mission Concept for The New Frontiers Program, *PSJ*, *in prep.*
- [5] **Pamerleau, I. F.**, Sori, M. M. (2026), Shallow floors, domes, and fractures on Callisto craters may be caused by deformation in an undifferentiated crust, *JGR: Planets*, *in revision*.
- [4] **Pamerleau, I. F.**, Reid, M., Rowland, J. C., Shelef, E. (2026), [Delineation of river flats with local hypsometry](#), *Geomorphology*, 513, 110504.
- [3] Blanco-Rojas, M., Sori, M. M., **Pamerleau, I. F.** (2026), [Peaking into Oberon's past: An isolated peak could imply the existence of a subsurface ocean or convecting ice shell](#), *JGR: Planets* 131, e2025JE009015.
- [2] **Pamerleau, I. F.**, Sori, M. M., Scully, J. E. C. (2026), [Asymmetric crater relaxation on an ice-rich Ceres driven by insolation](#), *JGR: Planets*, 131(3), e2025JE009374.
- [1] **Pamerleau, I. F.**, Sori, M. M., Scully, J. E. C. (2024), [An ancient and impure frozen ocean on Ceres implied by its ice-rich crust](#), *Nature Astronomy*, 8(11), 1373–1379 (won Pellas-Ryder Award for best student paper in planetary science, 2024).

Conference Abstracts

- [16] **Pamerleau, I. F.**, Sori, M. M. (2026), The Evolution of Mercury's Hollows via Topographic Diffusion, *AGU 2026*, #2071159.
- [15] **Pamerleau, I. F.**, Sori, M. M. (2026), Topographic Diffusion and evolution of Mercury hollows, *LPSC 57th*, 1264.
- [14] Miller, S. G., Alian, O. M., Borrelli, M., Christian, J. R., Detelich, C., Ferreira, J. P., Hon, O., Jhoti, E., Kubas, A. R., Lutz, K. A., **Pamerleau, I. F.**, Pickett, N. B., Rountree, R. D., Sacks, L. E., Thakar, B., Tuggle, J. P., Valantinas, A., Zomerdijk-Russell, S., Nash, A. E., Scully, J. E. C. (2025), A New Frontiers Class Mission Concept Study to Explore Triton, *AGU 2025*.
- [13] **Pamerleau, I. F.**, Sori, M. M. (2025), Insights into Callisto's interior: shallow craters and broad domes as a test for JUICE, *EPSC-DPS 2025*, 933.
- [12] **Pamerleau, I. F.**, Sori, M. M. (2025), Broad topographic domes as a test of Callisto's interior structure, *LPSC 56th*, 1329.
- [11] Blanco-Rojas, M., Sori, M.M., and **Pamerleau, I. F.** (2024), Constraining Oberon's thermal history from the evolution of one of the tallest peaks in the Solar System, *The Uranus Flagship: Investigating new paradigms for outer planet exploration*.

- [10] Sori, M.M., Blanco-Rojas, M., Bramson, A. M., Cartwright, R.J., Menten, S.M., Nordheim, T.A., and **Pamerleau, I. F.** (2024), Endogenic and exogenic evolution of the large Uranian moons can be revealed by observations of their surfaces from a Uranus Flagship mission, *The Uranus Flagship: Investigating new paradigms for outer planet exploration*.
- [9] **Pamerleau, I. F.**, Sori, M. M., Scully, J. E. C. (2024), Asymmetric Relaxation of Large Craters in an Ice-Rich Crust are Consistent with Dawn Observations of Ceres, *LPSC 55th*, 1263.
- [8] Blanco-Rojas, M., Sori, M. M., and **Pamerleau, I. F.** (2024), Oberon as an ocean world? Insights from the topography of one of the tallest peaks in the Solar System, *LPSC 55th*, 1468.
- [7] Sori, M. M. and **Pamerleau, I. F.** (2023), Thermal history of Uranian moons and their oceans from topography: Constraints from Voyager 2 and prospects for a Uranian orbiter, *Uranus flagship: Investigations and instruments for a cross-discipline science workshop*, 8072.
- [6] **Pamerleau, I. F.**, Sori, M. M., Scully, J. E. C. (2023), An ice-rich crust with unrelaxed craters on Ceres reflects an ancient frozen ocean, *LPSC 54th*, 1359.
- [5] **Pamerleau, I. F.**, Sori, M. M., Johnson, B. C. (2023), Convection in Callisto's Ice Shell: Implications for Differentiation, *LPSC 54th*, 1647.
- [4] Sori, M. M., Laferriere, K. L., Burkman, K.S., Herring, J., Klidas, A., Manelski, H.T., McGlasson, R. A., Menten, S.M., **Pamerleau, I. F.**, and PérezCortés, S.L. (2023), Hollows as a source for Mercury's polar organics, *LPSC 54th*, 1103.
- [3] Kring, D., Bamber, E., Blance, A., Brezfelder, J., Faucher, J., Flom, A., Lehman Franco, K., Harris, E., Jhoti, E., Laferriere, K. L., Martin, A., Meyer, M., **Pamerleau, I. F.**, Plan, A., Roberts, E., Shubham, S., Slumba, K., Zimmermann, N., Barrett, T., (2023) Cascading Boulder and Boulder Track Experiment at Barringer Meteorite Crater (AKA Meteor Crater), Arizona, *LPSC, 54th*, 2186.
- [2] **Pamerleau, I. F.**, Sori, M. M., Scully, J. E. C. (2022), Insolation-Driven Topographic Evolution on Ceres, *LPSC, 53rd*, 1711.
- [1] **Pamerleau, I. F.**, Reid, M., Shelef, E., Rowland, J. C., Schwenk, J., Mishra, U. (2020), Automated Mapping of Arctic Floodplains to Improve Estimates of Sediment and Carbon Fluxes, *AGU Fall Meeting*, # H137-0002.

Skills

COMSOL Multiphysics (Finite Element Software)
JMARS & QGIS (GIS Softwares)
MATLAB (Topotoolbox, Mapping Toolbox) & Python

References

Associate Professor Michael Sori

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Professor Brandon Johnson

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